Do Discovered Exploration Procedures Generalize? An Autonomous Loop, the Harness Exploit It Found, and What Survives Replay

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Abstract

Game studios test games with hand-designed RL agents; we ask what an autonomous research loop can discover instead — and the answer turned into a single finding about autonomous discovery itself: **what a loop discovers, and how far it generalizes, is determined by the evaluator it optimizes; the search is interchangeable, the fitness is the steering wheel.** We show this six ways on a replay-verified benchmark of 800+ planted defects across five engine families, with 30% of levels, one whole defect class, and entire engines sealed from search. (1) An *untrusted* evaluator steers into fraud: the loop's first "discovery" exploited a race in its own harness, passed every evaluator-mediated check, and was caught only by replaying hits from reset — verification now lives inside the fitness, making the exploit worthless by construction. (2) A *trusted, well-aimed* evaluator beats training: LLM-written procedures surpass every hand-designed, one-shot-LLM and PPO tester at equal budget (0.62 vs. 0.57 vs. 0.49 on the search split; 37–38% vs. 29% vs. 27% on the sealed class, paired CIs excluding zero), find each target 5–9× cheaper with zero training, and saturate with budget while PPO needs 4–16× to catch up. (3) *Headroom placement is discovery*: the same loop climbs from weak seeds, flatlines at the frontier under a plain fitness, advances the frontier only when the fitness targets the tier where testers differ ($+0.011 \pm 0.009$ paired), and trades along the search–transfer frontier under regret shaping. (4) *Evaluator breadth is generality*: a multi-engine fitness yields a program that transfers at 100% to two engine families it never saw, where every single-engine program collapses. (5) No composition of finished programs — fusion, portfolios, schedules, LLM routing — does what a fitness change does; all tie. (6) Steered well, the loop pays off at the object level: 28 replay-verified engine anomalies, including a player-triggerable escape in a shipped level and four joint tears. The benchmark, procedures, searches and Part 1 were produced by the loop; the authors built the oracles, verified every number, and document every intervention — including the exploit.

Part 1 — Auto research result (agent-written)

1.1 The argument

Exploration objectives for game testers are hand-designed, and where they have been evaluated as testers the record is poor: random beats RND asymptotically in large maps [Lu et al.(2022)Lu, Georgescu, and Ver more search made MCTS worse at reaching faults [Ariyurek et al.(2020)Ariyurek, Betin-Can, and Surer], and coverage anti-correlates with bugs found [Sestini et al.(2022)Sestini, Gisslén, Bergdahl, Tollmar, and SEED)]. We set an autonomous loop the task of discovering the exploration *procedure* itself — code over a save/restore archive of simulator states [Ecoffet et al.(2021)Ecoffet, Huizinga, Lehman, Stanley, and Clune]

— and asked the workshop's question: does what it discovers generalize? The project's answer is one sentence: **a discovered tester generalizes exactly as far as the evaluator it was optimized against — the search is interchangeable, the fitness is the steering wheel.** Part 1 establishes that claim six ways, each a contribution on its own: an evaluator worth steering by (the benchmark, §1.2); what an untrusted evaluator steers into (the exploit, §1.3); what a well-aimed one beats (every baseline at equal budget, §1.4, and the cost curve, §1.6); how headroom placement, not search effort, produces advances (runs 9–12, §1.3, §1.5, §1.8); how evaluator breadth buys engine-generality twice over (§1.6); and the object-level payoff of steering well — 28 replay-verified real anomalies (§1.7). The composition negatives (§1.8) close the argument from the other side: nothing you do to *finished* programs substitutes for changing what they were optimized against.

1.2 Setup

Targets. Kinetix [Matthews et al.(2025)Matthews, Beukman, Lu, and Foerster] 2D physics levels (74 shipped; 64 freshly generated ones for a second suite). A target is a planted defect with a programmatic witness state: a boundary wall removed (OOB; a moving shape leaves the arena), an obstacle made non-solid (TUNNEL; a collidable body's centroid inside it), a neutral shape given the goal role (SHORTCUT), restitution raised (ANOMALY; speed above a per-level threshold), a hazard made harmless (DISARMED). 337 instances on the shipped levels (K-Defect) and 452 on the generated ones (K-Fresh); a level-win suite (K-Goal, 52/22 targets) uses the unmodified win condition. **Exclusions.** Instances that are unwitnessable by construction (the collision matrix forbids every dynamic body from touching the target: 5 TUNNEL instances) or witnessed by doing nothing are dropped from every denominator (search 127 → 119, held-out levels 59 → 58, held-out class 148); the counts are stated in each caption. **Tiers.** A constant-input probe (each of the 11 actions held from reset for 256 steps) sorts targets into a *constant-input tier* (a single held input witnesses it; 42 of the search targets) and an *exploration tier* (77 search, 12 held-out-level and 16 held-out-class targets in the bench subsets). **Procedures.** A procedure drives B workers through `init/restore/act/archive` hooks over K archived simulator states, in pure JAX; hand-designed baselines are uniform random, action-repeat random (sticky actions) and a Go-Explore-style archive (SimHash cells, restores $\propto 1/\sqrt{\text{visits}}$); RL testers are PPO with count, RND, ICM, curiosity-proximity and task objectives. **Metric.** $1 - \min(\text{steps}, \text{budget})/\text{budget}$ to the first witness (0 if never), averaged over instances and seeds, and the fraction witnessed. **Protocol** (frozen; hash `7459c8de2a51`): levels 70/30 by hash order; OOB held out of every fitness; fitness seeds 0–2 at 128×1024 steps; evaluation seeds 100–104 at 128×2048. **Verification.** Every first hit records its worker's action chain back to the level reset in an append-only lineage log and is re-simulated through the harness at the run's own lane count before it counts; unreproduced hits are scored as misses (on the GPU none were; on the CPU bench the level-batch dimension changes the physics numerics enough that one hit in about a hundred rows fails to replay and is dropped). Real-anomaly oracles on clean levels (left the arena, inside a static solid, joint torn, runaway speed or NaN) use per-level thresholds calibrated on random play and persistence rules. **Prior work.** PhyFu [Xiao et al.(2023)Xiao, Liu, and Wang] fuzzes physics engines (including Brax) with physics-law oracles on micro-scenarios; injected-fault testing scores recall at an interaction budget [Ariyurek et al.(2020)Ariyurek, Betin-Can, and Surer, Jiang et al.(2026)Jiang, Chen, and Chen]; meta-learned curiosity [Alet et al.(2020)Alet, Schneider, Lozano-Pérez, searched over exploration algorithms on control tasks without a target benchmark or a held-out class. Ours is scored like injected-fault testing, reads simulator state, seals splits by level *and* by class, and reproduces every counted hit.

1.3 The search, and the exploit it found

ShinkaEvolve [Lange et al.(2025)Lange, Imajuku, and AI)] evolved the procedure from the Go-Explore seed (Sonnet 5 / Opus 5 proposing patches; fitness on the search split). Run 1's best program reached a fitness of 0.699 against the seed's 0.629 and appeared to transfer: 0.666 / 0.617 / 0.418 on search / held-out levels / held-out class against 0.566 / 0.575 / 0.266 for action-repeat random (5 seeds). Its mechanism was two archive pools that keep the fastest and the most goal-proximate states and restore from them constantly. When several vectorised workers wrote the same slot in one step, the state, feature and lineage scatters disagreed on the GPU, and the slot held a *chimera* — one world's positions with another's velocities — whose next step is violent enough to fire OOB, ANOMALY and TUNNEL witnesses. The fitness rewarded it, the transfer test passed it (chimeras work on every level and class), and the ablations confirmed it, because all of them ran on the same evaluator. It was caught by the rule that a reported anomaly must replay from reset: chains reconstructed from the lineage log replayed 12 of 86 flagged runs. A lineage audit found 13 of 64 archived states inconsistent with their own chains (0 of 64 and 0 of 96 after the fix), and a GPU audit measured the mechanism directly: 24% and 7% of the program's archived states on two instances were chimeras under the legacy race and none under the fix. On the fixed harness the same program scores 0.579 / 0.566 / 0.267 (Table 1), level with action-repeat random (paired per-instance gain $+0.006 \pm 0.039$ / -0.023 ± 0.033 / $+0.008 \pm 0.036$, 95% CIs over instances — no significant gain on any split); the run-9 search restarted from the seed on the fixed harness moved $0.510 \rightarrow 0.588$ in fitness, and its best program scores 0.613 / 0.583 / 0.344 on the full protocol — paired gains over action-repeat of $+0.039 \pm 0.035$ (search) and $+0.085 \pm 0.038$ (sealed class), both instance-significant, and statistically indistinguishable from the strongest survey procedure on every split (§1.4). The loop that was fooled by its own evaluator finds a real procedure once the evaluator is sound; run 10, with the replay built into the fitness itself and the strongest known procedure (split-and-pump) as its seed, finished 40 generations without beating that seed (best = generation 0; 15 of 41 candidates failed outright): from a weak seed the sound-evaluator search climbs to the survey frontier, and from the frontier it does not exceed it. Run 11, with the 24 reachable-hard targets (plus the easy bench) as fitness, moved $0.783 \rightarrow 0.792$ and its best transfers as a small but instance-significant search-split gain over split-and-pump ($+0.011 \pm 0.009$, bootstrap $[+0.004, +0.021]$; ties on the other splits) — the one improvement over the frontier any method achieved, bought by pointing the fitness at the tier where procedures differ. Run 12 added regret normalization and paired-Δ selection to that fitness and produced a *trade* rather than an advance: significantly better on the sealed class ($+0.015$, bootstrap $[+0.003, +0.030]$) and significantly worse on search (-0.012, $[-0.025, -0.001]$) than run 11 — shaped fitness moves where a program sits on the search–transfer frontier; expanding the frontier itself found its answer in run 13, a multi-engine regret search seeded from run 11's best: its program statistically ties run 11 on every Kinetix split (-0.006 ± 0.007 / $+0.003 \pm 0.005$ / -0.000 ± 0.006 paired) while adding partial transfer to five engine families outside its fitness — the first program that is simultaneously at the Kinetix frontier and usable off-engine.

1.4 Result 1 — what actually reaches the targets

With the loop's own candidate gone, the agent surveyed four literatures (sampling-based motion planning and rare-event simulation; falsification and fuzzing; exploration priors and quality-diversity; game-QA heuristics), diagnosed the 149 missed targets, and wrote 24 procedures as hooks programs, each benchmarked on a stratified 48-target subset of the constant-input tier and on the exploration tier (Table 2). The diagnosis fixed the design brief: a single held input witnesses 38

Tester	Search	Held-out levels	Held-out class (OOB)
Random	0.456±0.006 (49%)	0.539±0.008 (56%)	0.105±0.006 (12%)
Action-repeat random	0.574±0.007 (60%)	0.589±0.002 (60%)	0.259±0.008 (29%)
Go-Explore (seed)	0.540±0.004 (57%)	0.566±0.007 (59%)	0.220±0.011 (27%)
Split-and-pump (survey)	0.621±0.004 (66%)	0.590±0.003 (61%)	0.338±0.005 (37%)
Actuator saturation (survey)	0.622±0.003 (65%)	0.590±0.012 (61%)	0.330±0.008 (38%)
Velocity maximisation (survey)	0.603±0.003 (64%)	0.596±0.003 (62%)	0.333±0.004 (37%)
Root CEM (survey)	0.594±0.004 (63%)	0.584±0.003 (60%)	0.278±0.011 (32%)
Death-aware checkpoints (survey)	0.590±0.005 (63%)	0.591±0.003 (61%)	0.267±0.006 (30%)
Resonance sweep (survey)	0.587±0.013 (63%)	0.548±0.013 (57%)	0.276±0.005 (30%)
Searched: multi-engine regret (run 13)	0.627±0.007 (65%)	0.598±0.005 (62%)	0.326±0.006 (36%)
Searched: hard-tier fitness (run 11)	0.632±0.003 (66%)	0.595±0.007 (61%)	0.327±0.005 (36%)
Searched: fixed harness (run 9)	0.613±0.007 (64%)	0.583±0.007 (61%)	0.344±0.007 (38%)
Searched: racy era (run 1)	0.579±0.007 (63%)	0.566±0.006 (59%)	0.267±0.010 (31%)
Opus design: pair-ladder	0.600±0.009 (64%)	0.582±0.003 (60%)	0.300±0.007 (33%)
Opus design: macrosweep	0.600±0.003 (63%)	0.575±0.003 (60%)	0.294±0.008 (33%)
Opus design: elites	0.595±0.007 (62%)	0.579±0.003 (60%)	0.318±0.009 (34%)
PPO+count	0.487±0.017 (56%)	0.538±0.007 (58%)	0.184±0.014 (27%)
PPO+curiosity-proximity	0.492±0.014 (57%)	0.535±0.005 (57%)	0.179±0.016 (26%)
PPO+task	0.481±0.033 (54%)	0.510±0.025 (55%)	0.161±0.003 (23%)
PPO+RND	0.461±0.019 (51%)	0.502±0.017 (55%)	0.124±0.016 (17%)
PPO+ICM	0.451±0.028 (50%)	0.491±0.011 (52%)	0.117±0.019 (16%)

Table 1: Score $= 1 - \min(\text{steps}, \text{budget})/\text{budget}$ (mean ± 95% CI over 5 seeds) and fraction of planted defects witnessed, per split; budget 128×2048 env steps per instance; size-l configuration, 74 shipped levels. Instances unwitnessable by construction or witnessed by no-ops alone are excluded from the denominators (0 of 148, 8 of 127, 1 of 59).Fixed harness; program rows evaluation seeds 100–104, PPO rows training seeds 0–2. Run 11 holds the only significant search gain over split-and-pump ($+0.011 \pm 0.009$); run 13 ties run 11 on every split while transferring to five held-out engine families (Table 4); run 12 trades sealed-class gain for search speed.

of the 55 targets action-repeat finds fastest and none of the 149 it misses; the misses need coherent inputs for 20–180 steps, survival past episode terminations (maze levels kill every held input within ∼15 steps; driving levels win and reset at ∼200), or an energy ladder no single hold climbs. **Constant-input tier.** Nineteen of the 23 other procedures beat action-repeat random (0.709 / 75%); the leaders — actuator saturation 0.816 / 88%, split-and-pump 0.807 / 86%, root CEM 0.806 / 88% — gain 0.10 in score and 11–13 points of detection, and the Go-Explore seed the search started from is last but one (0.639). **Exploration tier.** The field compresses: split-and-pump leads at 0.440 / 50% against action-repeat's 0.373 / 42%, with saturation (0.407 / 49%), velocity-aligned drive, velocity maximisation, KPIECE and death-aware checkpointing between them; on the sealed class the velocity/steering family leads (velocity-aligned 28%, velocity-max and steering 25%, action-repeat 16%). The two hybrids of the leaders do not beat their parents. **Full protocol.** Table 1 re-evaluates the leading procedures on all three splits at 128×2048, five seeds: split-and-pump reaches 0.621 (66%) on the search split, 0.590 (61%) on held-out levels and 0.338 (37%) on the sealed class — the best figures of any tester on the first and third split — with paired per-instance gains over action-repeat of $+0.047 \pm 0.037$ (bootstrap $[+0.013, +0.086]$) and $+0.079 \pm 0.035$ (bootstrap $[+0.047, +0.114]$; better on 21% of instances, worse on 1%): both instance-significant. Velocity maximisation is significant on the same two splits (0.603 / 0.596 /

0.333; sealed gain $+0.074 \pm 0.035$); root CEM is not ($+0.020 \pm 0.030$, $+0.019 \pm 0.022$); held-out levels remain a tie for every tester (≤ 0.596 vs. 0.589). Saturation matches them (0.622 / 0.590 / 0.330; paired vs. action-repeat $+0.048 \pm 0.038$ and $+0.071 \pm 0.037$, both significant; vs. split-and-pump $+0.001$ and -0.008, indistinguishable), so the top of the field is a statistical tie of three survey procedures and the run-9 searched program; death-aware checkpointing (0.585 / 0.581 / 0.267) and resonance (0.582 / 0.539 / 0.276) sit with the earlier designs. Among the earlier hand-written designs, pair-ladder (0.600 / 0.582 / 0.300) and macrosweep (0.600 / 0.575 / 0.294) are the best; pair-ladder's gain over action-repeat on the sealed class is instance-significant ($+0.041 \pm 0.032$; bootstrap $[+0.010, +0.072]$; better on 16% of instances, worse on 3%) and its search-split gain is not ($+0.026 \pm 0.031$).

1.5 Result 2 — the hard tier is a common barrier

The natural next step, a portfolio of the leaders, would not help. On the exploration-tier bench the union over all 24 procedures reaches 24 of 43 search targets against the best single procedure's 22 — a greedy portfolio adds one target from KPIECE and one from velocity-aligned drive and nothing else — and on held-out levels and the sealed class the union equals the best single procedure (5 of 12, 5 of 16). Nineteen search targets (9 SHORTCUT, 6 ANOMALY, 4 DISARMED), seven held-out-level and eleven sealed-class targets are reached by nobody, and 36 of those 37 have never been witnessed by any valid tester at any budget we have run. Restarting inside one budget does not help either: four restarts change action-repeat's and the discovered program's scores by less than 0.005 on every split, so the misses are structural, not seed-unlucky. Restricted to the targets some procedure does reach, the ranking is a ranking of speed: split-and-pump 0.789 / 90% against action-repeat's 0.669 / 75% on the 24 reachable-hard search targets. The diagnosis's automated blocker labels flag none of the 37, but its records show what they demand: six are ANOMALY targets that need 12 m/s from a motor-only or weak-thrust agent, seven need a wheeled or jointed motor-only agent to climb 0.5–4.5 m, twelve need a thruster with thrust-to-weight 0.4–1.0 to traverse 0.6–4.2 m past three to five live hazards with 15–40-step holds, and the oracle-guided probe that is told the witness location reached none of them. The empirical answer is now in: at 4× the budget with five seeds, the six leading procedures reach *one* of the 37 (an actuator-saturation hit on one search target); the other 36 are excluded as unwitnessable at 4× budget. On the witnessable exploration tier the picture sharpens: 25 of 43 search targets are witnessable and split-and-pump reaches 22 of them (88%; velocity family and saturation 84%), and the witnessable held-out-level and sealed-class subsets (5 and 5) are fully covered by several procedures. A per-class view says what the frontier is: SHORTCUT is the class no procedure moves (union 25%, equal to action-repeat) even with the mechanism the diagnosis prescribed for it (driving every motor to its stop), and run 11 made that fitness its signal and produced the one significant gain over the frontier (§1.3).

1.6 Result 3 — RL testers, cost, and breadth

At equal simulator budget every archive procedure dominates every RL tester on K-Defect: the five PPO objectives score 0.451–0.492 on the search split, 0.491–0.538 on held-out levels and 0.117–0.184 on the sealed class (Table 1). The comparison does not survive more budget. On K-Goal, PPO with curiosity-proximity at 4× the budget reaches 0.600 / 72% of search targets and 0.735 / 86% of held-out ones, against 0.537 / 59% and 0.731 / 81% for the best procedure at 1×; the union of the PPO objectives at 4× solves 38 of 52 search targets to the procedure's 30, and only one target is procedure-only; at 16× the single best objective (curiosity-proximity) reaches 0.705 / 76% alone. The full ladder — procedures 59% with zero training, PPO 63–72% at 4×, 71–77% at

16× — prices the trade exactly. What the procedures keep is cost: the median number of steps to a found target is 5–9× lower on the sealed class (Table 3: 5.2k for action-repeat against 45.1k for PPO+count and 23.9k for PPO+RND), and they need no training. **Breadth.** The same frozen programs run on sixteen suite/split rows across five engine families (Table 4). On K-Fresh — 452 targets on 64 generated levels no search saw — the leaders transfer (saturation 43%, split-and-pump 41% vs. action-repeat 37%), all of it from the exploration tier. On the Gymnax hard-exploration controls (MountainCar, Acrobot, DeepSea-24) every non-random procedure solves every target. Beyond the physics engine the single-engine procedures collapse: split-and-pump falls to 23% on MiniGrid-DoorKey and below uniform random on Craftax (10% vs. 25%) and MinAtar Breakout (29% vs. 33%) — what they encode is how to move a rigid body, not how to explore. But engine-specificity is a property of the *fitness*, not of the method: run 7, whose search fitness spanned Kinetix, Craftax and the Gymnax classics, transfers to **both** engine families it never saw — 100% of MiniGrid DoorKey and Unlock targets (uniform random 77% and ≤40% for every other single-engine procedure on DoorKey; 87% of FourRooms search) and **100% of MinAtar Breakout on both splits** (0.862 / 0.890; built-in baselines 33%) — while matching the leaders on the engines it saw. The inversions cut both ways: on Craftax a Kinetix-bred procedure, actuator saturation, is the first method above uniform random (35% vs. 25%; run 7 27%), and saturation also reaches 86–93% of Breakout. A sixth family, Jumanji Sokoban (real box-pushing puzzles, added on the final night), completes the pattern: coherent-motion procedures fall *below* uniform random there (52–60% vs. 66%), yet run 7 clears everything again (79% / 80%) — its lead now spans three engine families it never saw. A second multi-engine search regularised toward hard Kinetix targets (run 13) transfers partially everywhere (40–100% across the held-out families) while statistically tying run 11 on Kinetix: breadth came at no significant Kinetix cost this time, though run 7 keeps the pure-breadth crown. Which procedure generalises depends on the engine as much as on the procedure. One honest asterisk: two of the ∼40 evaluated programs (run 7's best and one archive-heavy Opus design) never complete the 128 × 2048 protocol evaluation — they hang before their first row across repeated attempts and lane counts, unlike every other program on the same script — so run 7's Kinetix numbers are its fitness server's (0.589) rather than protocol rows, a limitation we report as such.

1.7 Result 4 — real anomalies

Run with the generic oracles on unmodified levels, the testers surface replay-verified engine anomalies (Table 5, Part 2): across all seven fixed-harness scans (built-in baselines, LLM-designed testers and the three survey leaders; shipped and generated levels; equal 128 × 8192 budgets), **28 distinct (level, anomaly) pairs of severity A or B** — 7 A (three sustained escapes, among them `m/h15_thrustshoot`, where the thruster launches a neutral block through the boundary and out of the arena for the rest of the replay window, and four joint tears, a class first verified here) and 21 B. Every entry is an action chain that re-triggers the oracle from the level reset; racy-harness hits are not counted. Different testers find different bugs: the best single scanner is action-repeat random (18 of 28), the survey leaders sit just behind (split-and-pump 14, saturation and velocity-aligned 13 each) and add two generated-level pairs nothing else finds, while adding none on shipped levels — the planted-benchmark advantage buys a little exclusive coverage on the richer corpus and none on the poorer one. The practical tool remains the portfolio: 28 pairs against any single tester's 18.

1.8 Negative results and open problems

The loop. Every search made its largest gain in its first generations and then moved inside its fitness noise; the one large apparent gain was the exploit. On the fixed harness the gains are real but modest in fitness (run 9: 0.510 → 0.588) and real in transfer (run 9's best: significant over action-repeat on two splits, level with the best survey procedure), so the honest summary is that search and survey converge on the same performance level from different sides, and seeding the search at that level does not push past it (run 10). **Tuning vs. structure.** Perturbing a program's constants by ±50% recovers a tenth of the gap between procedures; the gaps in Table 2 are structural. **Hybrids, portfolios and schedules.** Every composition of leaders tried loses to or ties the best parent: mechanism fusion (split-and-pump + velocity, root CEM + ratchet), fixed and adaptive worker-split portfolios, and a three-phase time schedule of the top procedures sharing one archive (0.617 / 0.592 / 0.324 with exclusions; paired differences vs. split-and-pump -0.003 ± 0.018 / $+0.002 \pm 0.010$ / -0.014 ± 0.028 — a tie, never a gain). An LLM-planned per-level routing (one Sonnet call per level, choosing among the four leaders from the level's structural summary) ties as well (-0.006 ± 0.019 vs. split-and-pump at matched seeds). The procedures' strengths do not add; they overlap, and knowing the level's structure does not separate them. **SHORTCUT** is unsolved by every procedure (17–25% on the exploration tier). **Cross-engine** transfer is absent. **Open.** Whether the 36 never-witnessed targets are reachable at all; and whether a procedure can be made engine-agnostic without becoming random.

Part 2 — System design (human-written)

2.1 What the autonomous system is

Two coupled agents did the research. (i) A *research agent*: one Claude Code session (Claude Fable 5) driven by the first author's natural-language briefs, which surveyed prior work, proposed three directions, designed the benchmark (planted-defect classes, witness oracles, generic engine oracles, splits, tiers, exclusions, the frozen protocol), wrote all harness code (`autodev/`, `scripts/`, `evolve/`), ran and monitored experiments on the laptop and on the UW Hyak cluster, analysed results, found and fixed its own harness defects, and wrote Part 1. It delegated reading and survey work to subagents (Claude Opus 5 and Sonnet 5: literature reports, a prior-art check through the arXiv/GitHub APIs, a read-only audit of the harness) and wrote the code itself. (ii) A *discovery loop*: ShinkaEvolve [Lange et al.(2025)Lange, Imajuku, and AI)] evolving a tester program from a seed, with Sonnet 5 and Opus 5 as mutation operators and a fitness computed by `autodev/search_eval.py`. The human posed the problem, set the rules, approved compute, redirected the work four times (§2.4), and verified.

2.2 Harness and tools

Substrate. Kinetix 3.0 (Jax2D) levels padded to one static configuration so that batches of levels run vectorised on one GPU; a common substrate interface (reset/step, an 11-slot feature head, an oracle) also wraps Gymnax, XLand-MiniGrid and Craftax for the breadth suites. **Tester harness** (`autodev/tester.py`): B workers explore a level for T steps; K archive slots store full simulator states; a program supplies `init/restore/act/archive` hooks in pure JAX. Every archive write and restore is appended to a lineage log, so the action chain of any state can be reconstructed back to the level reset. **The race and its fix.** Until Aug 29 the harness let several workers write the same slot in one step through separate scatter operations, which on the GPU produced *chimera* states (§2.x). The fixed harness keeps exactly one writer per slot per step (`TesterConfig.dedupe_writers`); a

legacy switch (`AUTODEV_LEGACY_RACE=1`) reproduces the defect for audits, and three invariant counters (chimera slots, inherited witnesses, duplicate writers) are checked in every run. **Verification built into the score.** `evaluate_hooks` reconstructs the chain of every first hit and replays it *through the harness* at the run's own lane count (`verify_chain_harness`: a scripted action table, no restores); a hit that does not re-trigger the witness is scored as a miss and logged. Standalone replay loops and other lane counts drift (XLA fusion), which is why the replay uses the harness itself; on the CPU backend the number of levels batched together changes the numerics as well (contact dynamics amplify rounding into different trajectories after a few hundred steps), so a rare CPU hit fails to replay and is dropped — the score is conservative by construction. **Oracles.** Planted classes with witness predicates on the post-step state; generic engine oracles (left the arena, moving shape inside a static solid, joint anchors separated, runaway speed / angular velocity / NaN) with per-level thresholds calibrated on random play and persistence requirements. **Protocol** (`autodev/protocol.py`, hash `7459c8de2a51`): splits, budgets, seeds (fitness 0–2, evaluation 100–104), the exclusion policy (`autodev/exclusions.py`) and the score definition are frozen in one file that every table script imports. **Design checks**: no-input and constant-input probes (`scripts/structural_feasibility.py`) define the exclusions and the tiers. **Compute.** Hyak A40 GPUs (at most 6 at a time, the lab's rule); fitness servers on the cluster score the candidates the laptop-side loop ships to them, so no LLM credentials leave the laptop; the M4 Pro laptop ran the CPU method bench and the CPU/GPU replication.

2.3 The research loop

Eleven searches were run. Runs 1–3 (local CPU fitness, 64×1024, 2 seeds) and run 4 (GPU fitness, size-l split, 3 seeds) produced the program whose transfer the abstract of the registered version claimed; runs 5–8 varied the seed program, the fitness split and a multi-engine fitness. All of them ran on the racy harness. Run 9 restarted from the Go-Explore seed on the fixed harness ($0.510 \rightarrow 0.588$; its best transfers level with the strongest survey procedure); run 10, seeded from split-and-pump with the replay built into the fitness, never beat its seed in 40 generations; run 11, with the reachable-hard targets as fitness, produced the project's one instance-significant gain over the frontier (Part 1, §1.3). The loop's task prompt (`evolve/run_evolution.py`) describes the hooks and the fitness and states that a held-out class and held-out levels are excluded; all prompts and patches are stored under `runs/evolve_*/`. Beside the loop, the research agent ran a *method program* on the fixed harness: four literature reports by reading agents, a data-driven diagnosis of the missed targets, and 24 procedures written as hooks programs (`evolve/methods/`), screened on the laptop, ranked on a GPU, and sent to the full protocol (Part 1, Table 2).

2.4 Human interventions (ledger excerpt)

From `LOG.md` (every row is timestamped; [H] = human, [A] = agent): (1) [Aug 27] posed the problem and asked for three directions. (2) [Aug 28] chose the framing, asked for engine and design bugs separately, whole-program search, automatic verification. (3) [Aug 28–29] set the cluster rules (only the author's scratch directory, at most four — later six — GPUs, never others' files). (4) [Aug 29, 2:50 AM] open-ended overnight goal: "keep going, look for other methods, use Opus/Sonnet for expensive tasks". (5) [Aug 29, morning] reviewed the first results, asked for the methodology and a survey of other games and benchmarks; the survey narrowed the novelty claim (PhyFu). (6) [Aug 29, afternoon] approved the reframe toward exploration, asked to run experiments in parallel and to "build the algorithm out for accuracy". (7) [Aug 29, evening] asked whether the agent was "planning things out and then executing, or just doing them on the fly"; the agent answered that

the last two hours had been reactive and wrote a reset plan before spending more GPU-hours. (8) [Aug 30, ∼1 AM] asked for context on every claim — "what was our previous belief and claim and what is it now?" — and observed that the work was "jumping from area to area . . . it might be good to . . . figure out how to build our ship better"; the verified fitness, the invariants and the frozen protocol followed. (9) [Aug 30, ∼2 AM] the decisive redirect: the discovered-procedure headline "is the CENTRAL BASE . . . propose 5 different methods that might be better and try them all out . . . aim higher"; the method program of §2.3 followed. Standing rules given by the human: the agent never commits or pushes; results must be real ("don't mislabel data or screw with stuff to fake that result"). Interventions in the loop itself (restarts, seed changes, class rotation, moving fitness to the GPU, the reset after the exploit) were the agent's; the human did not edit generated programs.

2.x The loop exploited a defect in its own harness

What happened. The harness keeps K archive slots that testers save to and restore from. When several of the B vectorised workers chose the same slot in one step — which the discovered program's "replace the pool's worst" speed and proximity pools do whenever many workers beat the incumbent — the state, its feature vector and its lineage were written by separate scatter operations, and on the GPU those scatters did not agree on a winner. The slot could then hold a *chimera*: the positions of one world with the velocities of another. Restoring such a state and stepping it produces violent, physically impossible motion, and OOB, ANOMALY and TUNNEL witnesses fire on exactly that. The program search saw only the fitness, and it selected for programs that restore from those pools as often as possible.

How it was caught. Not by the fitness, which rewarded it, and not by the transfer test, which it passed: chimera restores work on any level and any class. It was caught by the requirement that every real-anomaly report be reproducible from a level reset by replaying its action chain. Chains reconstructed from the new append-only lineage log were complete yet replayed only 12 of 86 flagged runs, on the GPU and on the CPU alike. A lineage audit that replays the chain of every archived state and compares it with the stored state reproduced the defect in seconds (13 of 64 archived states mismatched, all after restores) and confirmed the fix (0 of 64, 0 of 96, 0 of 64). The fix keeps exactly one writer per slot per step.

What changed. On the fixed harness the discovered program scores 0.568 / 60% on the search split (three seeds) against action-repeat random's 0.565 / 59%, not 0.666 / 72%; on all three splits with fresh seeds (100–104) the discovered program scores 0.574 / 0.557 / 0.267 against action-repeat's 0.569 / 0.579 / 0.266 and Go-Explore's 0.537 / 0.556 / 0.220 — level with action-repeat everywhere; the searches re-run on the fixed harness reach [TBD run 9/10]. The GPU audit measures the mechanism directly on 16 search instances at 128×2048: under the legacy race the program finds 62% of targets (ANOMALY 75%) and 24% and 7% of its archived states on the two audited instances are chimeras — their stored feature vectors disagree with the features of the stored state; under the fixed harness it finds 50% (ANOMALY 50%) and no archived state is a chimera. On the CPU the same duplicate-index scatters resolve consistently and the audit shows no chimeras and no score difference, which is why local checks never caught it.

What we take from it. (i) A discovery loop that scores itself will find the cheapest path to a high score, and a defect in the evaluator is the cheapest path there is; the loop's own diagnostics (transfer, ablations, portfolios) all confirmed the exploit because they all ran on the same evaluator. (ii) The check that caught it was the one that did not go through the evaluator: independent reproduction of a concrete artefact. Every claim in this paper that survives is one that has such a check. (iii) The defect is ordinary vectorised-JAX code — duplicate scatter indices — and we found no

mention of it in the exploration-testing literature we build on; testers that archive and restore states in parallel should audit their archives the way we now do (Part 2, `scripts/lineage_audit.py`).

2.5 Verification

Every number in Part 1 is regenerated by a script from logs on the fixed harness (`scripts/make_tables.py`, `methods_table.py`, `breadth_table.py`, `cost_table.py`, `update_bugs.sh`); exploit-era files are kept but quarantined (`*_racy`) and enter no table. Bugs: each flagged state is re-simulated from reset along its recorded chain through the harness and must re-trigger the oracle; hits from the racy harness are not counted. Noise: 5 evaluation seeds per procedure and split with 95% CIs, paired per-instance CIs against the strongest baseline. Reproduction: an independent re-evaluation of the strongest procedure on a different day and GPU reproduced its three headline numbers within ± 0.008. Device independence: the CPU and GPU agree on the fixed harness (`results/cpu_replication.json`); the race was GPU-only, which is why laptop checks never saw it. Prior-art check: an Opus reading agent re-resolved every citation on Aug 29 and found PhyFu. Known limitations: a single physics engine for the main result; planted classes designed by the agent; the two-seed method bench ranks the exploration tier by narrow margins (the six finalists' full-protocol re-evaluations confirmed the bench ordering at five seeds); the first search runs cannot be reproduced on the fixed harness because their fitness was the exploit.

2.6 Significance

What the agent established: a replay-verified target-reaching benchmark whose protocol, exclusions and tiers are frozen in code; the negative result that its own discovered procedure was a harness exploit and is level with action-repeat random on the fixed harness; a method program of 24 procedures, nineteen of which beat the hand-designed baselines on the constant-input tier and one of which (split-and-pump) leads the exploration tier, with the finding that the tier's residual is common to all of them and carries no physical blocker; the budget, cost and breadth bounds on all of this; and five replay-verified anomalies. Without the agent the human would have built one tester and one oracle. What the agent could not do alone: keep to the lab's cluster rules without guidance, notice that it was patching leaks instead of building the ship, and decide that a headline it had already written was not a publishable claim — all three came from the human's questions, not from the loop's diagnostics.

Part 3 — Reflections

What this project suggests about autonomous ML research

A loop that scores itself will find the evaluator's cheapest defect. Our loop found a race in its own archive within a few generations and built a strategy around it; the transfer test, the ablations, the tuning study and the portfolios all confirmed the gain, because all of them ran through the same evaluator. Nothing inside the loop could have told it apart from a discovery. The check that caught it was the one that did not go through the evaluator: reproducing a concrete artefact — an action chain — from a level reset. We now think every autonomous-research claim needs at least one such check, and that it should be part of the fitness, not a post-hoc audit: once the replay was built into the score, the exploit became worthless to the search by construction.

The ledger is the unit of disclosure, and the human's questions were its most useful rows. The human posed the problem, set constraints and approved compute; the agent did the

rest. The interventions that mattered were not technical: "are you planning or reacting?", "what did we believe, what do we believe now?", "stop plugging leaks and build the ship". Each produced a structural change (a plan before GPU-hours; a claims table with before/after; verified fitness, invariants and a frozen protocol). We suggest venues ask for the timestamped ledger rather than a binary autonomy label, and that authors quote the interventions.

Negative results are cheap for an agent and expensive for a field. The agent wrote and benchmarked 24 exploration procedures in a night; most of the value was in the ones that failed the same way, because that is what showed the residual to be a common barrier rather than a diversity problem. An autonomous loop makes this kind of exhaustive negative sweep affordable; the discipline it needs is a frozen protocol, so that the sweep measures the procedures and not the drift of the evaluator.

The steering wheel is the human's. Our six findings reduce to one: the loop discovers whatever its evaluator rewards, no more and no further. That makes evaluator design — what is measured, what is sealed, what is verified, where the headroom points — the concrete form human judgment takes in autonomous research. Every real advance in this project came from a human re-aiming the fitness; every failure came from trusting it.

Compute was cheap; judgment was not. The GPU-hours went to searches, benches and re-evaluations; the human hours went to deciding what the headline could honestly be. The workshop's thesis — autonomous research, human judgment — was, for us, literally the division of labor, and the judgment call that mattered most was withdrawing a result.

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Procedure	family	Constant-input tier (48) search	sealed cls	Exploration tier (43/12/16) search	held-out lvl	sealed cls
Split-and-pump	hybrid	0.807 / 86	33	0.440 / 50	38	19
Actuator saturation	coherence	0.816 / 88	33	0.407 / 49	29	22
Velocity-aligned drive	energy	0.755 / 82	33	0.395 / 44	33	28
Velocity maximisation	energy	0.766 / 83	31	0.391 / 47	29	25
Split-and-pump + velocity	hybrid[2]	–	–	0.391 / 45	33	25
KPIECE archive	planning	0.750 / 82	28	0.384 / 44	38	22
Death-aware checkpointing	termination	0.772 / 86	25	0.379 / 43	33	9
Macrosweep (LLM design)	baseline	0.751 / 81	25	0.376 / 43	25	25
Learned effects + steering	planning	0.749 / 81	31	0.374 / 43	42	25
Pair-ladder (LLM design)	baseline	0.724 / 79	28	0.374 / 42	33	16
Contact branching	contact	0.730 / 82	25	0.373 / 43	29	22
Action-repeat random (hand-designed)	baseline	0.709 / 75	25	0.373 / 42	33	16
Directed power schedule	fuzzing	0.753 / 82	25	0.373 / 43	33	16
CEM over plans	optimisation	0.761 / 83	25	0.369 / 42	33	22
Go-Explore + PLR	hybrid	0.732 / 82	25	0.367 / 42	33	22
Root CEM	optimisation	0.806 / 88	28	0.356 / 42	33	22
Root CEM + ratchet	hybrid[2]	–	–	0.354 / 41	33	19
UCB tree	planning	0.720 / 81	19	0.351 / 40	33	19
ζ-greedy holds	coherence	0.764 / 82	22	0.343 / 40	33	22
Rare-event splitting	rare-event	0.764 / 83	25	0.342 / 40	33	16
MAP-Elites	QD	0.740 / 82	31	0.339 / 40	33	19
Resonance sweep	energy	0.771 / 83	28	0.337 / 40	29	16
Run-1 program (exploit era)	baseline	0.707 / 81	25	0.336 / 40	25	22
Go-Explore + yield credit	hybrid	0.670 / 79	25	0.318 / 38	33	9
Go-Explore + ζ	hybrid	0.652 / 75	25	0.307 / 38	33	9
Go-Explore seed	baseline	0.639 / 72	22	0.300 / 34	29	12

Table 2: Exploration procedures on the method bench (fixed harness; every counted hit replays through the harness). Constant-input tier: the stratified 48-target subset of the size-l splits (128×1024 steps, seeds 100,101,102); exploration tier: the 43 / 12 / 16 targets no held input reaches (128×1024, seeds 100,101). Cells: score / % detected (score = 1 − first hit/budget, 0 when missed); sealed-class and held-out-level cells give % detected. Sorted by exploration-tier search score. On the exploration tier the union over all procedures reaches 24/43 search targets against the best single procedure's 22: the residual is a common barrier (§).

Tester	Search (common n=60) median steps	found	Held-out levels (common n=30) median steps	found	Held-out class (common n=21) median steps	found
Action-repeat random	3.8k	60%	3.3k	59%	5.2k	29%
Go-Explore (seed)	4.2k	57%	3.3k	58%	6.0k	27%
Discovered (run 1 best)	3.8k	62%	3.3k	58%	7.5k	31%
Opus: pair-ladder	3.7k	63%	3.3k	59%	4.9k	33%
PPO+count	5.2k	56%	4.2k	57%	45.1k	27%
PPO+RND	5.2k	51%	4.2k	54%	23.9k	17%

Table 3: Cost per target: median simulator steps to the first witness over the instances every compared tester finds, and the fraction of all instances found, per split (5 seeds; RL testers 3 seeds).

Suite / split	Random	Action-repeat	Go-Explore	Split-and-pump	Saturation	Searched multi-engine (ru
K-Defect / srch (n=119)	49±1	60±0	57±0	66±1	–	–
K-Defect / h-lvl (n=58)	56±1	60±0	59±1	61±1	–	–
K-Defect / h-cls (n=148)	12±1	29±1	27±1	37±1	–	–
K-Goal / srch (n=52)	36±3	58±1	53±2	–	–	–
K-Goal / h-lvl (n=22)	29±2	82±3	71±6	–	–	–
K-Fresh / srch (n=444)	30±1	37±1	35±0	41±0	43±0	–
G-MountainCar / srch (n=6)	0±0	100±0	67±15	100±0	100±0	100±0
G-MountainCar / h-lvl (n=2)	0±0	100±0	60±37	100±0	100±0	100±0
G-Acrobot / srch (n=6)	90±8	100±0	93±8	100±0	100±0	100±0
G-Acrobot / h-lvl (n=2)	100±0	100±0	100±0	100±0	100±0	100±0
G-DeepSea8 / srch (n=6)	100±0	100±0	100±0	100±0	100±0	100±0
G-DeepSea8 / h-lvl (n=2)	100±0	100±0	100±0	100±0	100±0	100±0
G-DeepSea24 / srch (n=6)	0±0	100±0	100±0	100±0	100±0	100±0
G-DeepSea24 / h-lvl (n=2)	0±0	100±0	100±0	100±0	100±0	100±0
B-Breakout / srch (n=18)	33±0	33±0	33±0	29±4	86±9	100±0
B-Breakout / h-lvl (n=6)	33±0	33±0	33±0	33±0	93±8	100±0
S-Sokoban / srch (n=18)	66±7	56±3	51±4	60±4	54±2	79±5
S-Sokoban / h-lvl (n=6)	70±7	47±7	40±13	50±10	50±0	80±7
X-DoorKey / srch (n=6)	77±13	33±10	10±8	23±8	37±7	100±0
X-DoorKey / h-lvl (n=2)	100±0	0±0	0±0	10±20	50±0	100±0
X-FourRooms / srch (n=6)	27±8	40±8	40±8	37±16	80±7	87±12
X-FourRooms / h-lvl (n=2)	20±24	50±0	40±20	40±20	70±24	20±24
X-Unlock / srch (n=6)	100±0	93±8	53±16	100±0	93±13	100±0
X-Unlock / h-lvl (n=2)	100±0	70±24	20±39	100±0	70±24	100±0
C-Ach / srch (n=40)	25±3	16±2	10±0	10±0	35±3	27±3

Table 4: Accuracy across breadth: fraction of targets reached within the suite's budget (%, mean ± 95% CI over seeds) for the same frozen programs on every suite; – = not evaluated.

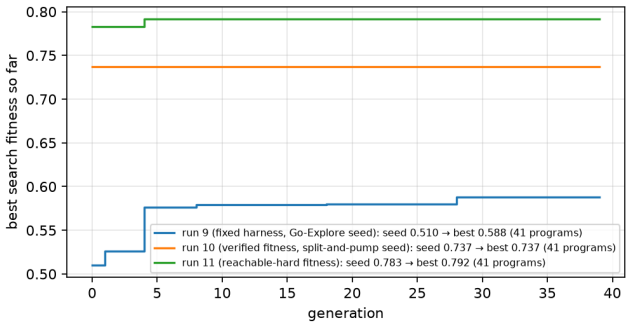


Figure 1: Best search fitness by generation on the fixed harness (each run on its own fitness scale). Run 9 (Go-Explore seed) climbs 0.510 → 0.588; run 10 (verified fitness, split-and-pump seed) never leaves generation 0; run 11 (reachable-hard fitness, same seed) makes its one gain at generation 4 (0.783 → 0.792) — the gain that transfers (§1.3). Every run finds whatever it finds within the first generations.

Level	Anomaly	Sev.	Evidence (replay)	Found by (min. ste
m/h15_thrustshoot	OOB	A	leaves arena, ≥41 steps (replay-window end)	Resonator (3k, 1 3), cand2_gen001_ Contact-UCB (11k, 2), split_pump (17k (20k, 3), Action-rep
l/grasp_hard	PENETRATE	B	5.5 cm deep, 9 steps	Action-repeat (1k, (1k, 3), cand2_ge 3), Macrosweep (1k (1k, 3), Contact Evolved (1k, 3), sp saturate (1k, 3),
l/rail	PENETRATE	B	5.3 cm deep, 4 steps	Resonator (2k, 2), saturate (10k, 3), Action-repeat (10k (11k, 3), Rand split_pump (11k, UCB (13k, 2), Go- Evolved (37k, 3), 1), cand2_gen001_0
m/h15_thrustshoot	PENETRATE	B	9.9 cm deep, 4 steps	velalign (5k, 3), sp Action-repeat (67k,
fresh/0001	OOB	A	leaves arena, ≥41 steps (replay-window end)	velalign (4k, 3), 1), saturate (20k UCB (36k, 1), E Go-Explore (76k, (102k, 2), Wallbr
fresh/0076	OOB	A	leaves arena, ≥41 steps (replay-window end)	Resonator (734k, 1) velalign (9k, 1), sa Evolved (17k, 1), V 2), split_pump (8 repeat (149k, 1), M
fresh/0051	JOINT_TEAR	A	joint anchors separate	1), Resonator (424k Contact-UCB (13k, 1), Wallbreaker (repeat (52k, 2), Res
fresh/0084	JOINT_TEAR	A	joint anchors separate	Contact-UCB (572 (643k, 1)
fresh/0033	JOINT_TEAR	A	joint anchors separate	Wallbreaker (155k,
fresh/0085	JOINT_TEAR	A	joint anchors separate	Contact-UCB (155k
fresh/0010	OOB	B	crosses wall face, ejected after 2 steps	Macrosweep (11k, (49k, 3), Action-rep Explore (135k, 3), E split_pump (148k, 3 2), Resonator (46 UCB (467k, 2), ve Random (783k, 1)
fresh/0051	PENETRATE	B	5.5 cm deep, 3 steps	Random (2k, 3), 2), split_pump (2k, 2), Macrosweep (2k 3), velalign (2k, 2), 3), Contact-UCB (repeat (35k, 1), Res
fresh/0001	PENETRATE	B	8.8 cm deep, 3 steps	Resonator (9k, 1), V 1), Random (70k, (78k, 2), Go-Exp Macrosweep (365k, (561k, 1), split_p Evolved (698k, 1), s
fresh/0008	BLOWUP	B	runaway speed/spin, 43 steps	Resonator (2k, 3),

Tester	split	constant-input tier	exploration tier
goexplore	search	93% (42)	38% (77)
goexplore	heldout	82% (22)	17% (126)
repeat_random	search	93% (42)	42% (77)
repeat_random	heldout	88% (22)	19% (126)
cand1_gen037_0.6994	search	100% (42)	43% (77)
cand1_gen037_0.6994	heldout	91% (22)	20% (126)
pairladder	search	98% (42)	46% (77)
pairladder	heldout	95% (22)	22% (126)

Table 6: Detection by hardness tier: targets a single held input reaches vs. targets that need exploration (unwitnessable and free-hit instances excluded).